

Department of Computer Science & Engineering

A Industry Oriented Mini Project Presentation
on

DEEP LEARNING-BASED INSURANCE CLAIM VERIFICATION USING IMAGE ANALYSIS FOR FRAUD DETECTION

Batch No: E14

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ABSTRACT

Insurance fraud is a significant challenge for the insurance industry, leading to substantial financial losses and increased premiums for genuine customers. This project proposes an intelligent system for identifying fraudulent insurance claims using advanced image verification techniques. The system leverages deep learning models, particularly object detection and image classification algorithms such as YOLOv8 and convolutional neural networks, to analyze images submitted with claims. It evaluates visual evidence to detect inconsistencies between reported damages and actual conditions, identifies signs of image tampering, and assesses damage authenticity. Additionally, the system integrates metadata analysis and feature extraction to enhance decision-making accuracy. A fraud detection engine combines multiple indicators, including object mismatch, damage inconsistency, and anomaly detection, to classify claims as genuine or fraudulent. The proposed framework is implemented using a Flask-based web application that allows users to upload claim images and receive real-time analysis results. By automating the verification process, the system improves efficiency, reduces manual inspection efforts, and enhances fraud detection accuracy. This approach contributes to building a more reliable, scalable, and intelligent insurance claim processing system.

1. INTRODUCTION

- ❖ Insurance fraud causes major financial losses for companies and increases premium costs for honest customers.
- ❖ Traditional claim verification methods are time-consuming, manual, and often unable to detect sophisticated fraud.
- ❖ This project introduces an intelligent fraud detection system that uses image verification and deep learning techniques.
- ❖ Advanced models such as YOLOv8 and Convolutional Neural Networks (CNNs) analyze claim images to identify fake, tampered, or inconsistent damage.
- ❖ The system also examines image metadata, feature patterns, and anomalies to improve fraud detection accuracy.
- ❖ A Flask-based web application provides real-time claim analysis, helping insurers process claims faster and more reliably.
- ❖ The proposed solution aims to reduce fraud, save costs, and create a smarter insurance claim verification process.

2. Objectives of the Project

- ❖ To develop an intelligent system for detecting fraudulent insurance claims using image verification techniques.
- ❖ To use deep learning models such as YOLOv8 and CNNs for analyzing claim images and identifying damaged objects.
- ❖ To detect inconsistencies between reported damages and actual visual evidence.
- ❖ To identify image tampering, fake damages, and suspicious claim patterns.
- ❖ To analyze image metadata and extract important features for improving fraud detection accuracy.
- ❖ To combine object detection, anomaly detection, and classification methods into a single fraud detection engine.
- ❖ To provide real-time claim analysis through a Flask-based web application.
- ❖ To reduce manual inspection efforts, save processing time, and improve decision-making efficiency.
- ❖ To enhance the reliability, scalability, and accuracy of insurance claim verification systems.

4. SYSTEM ANALYSIS

- **Existing System**

- Traditional insurance claim verification mainly depends on manual inspection and human judgment.
- Claim officers review submitted images and documents to identify damage authenticity.
- Fraud detection is based on rule-based checks and past claim history.
- Manual processes are slow, costly, and prone to human error.
- Detecting image tampering and fake damage evidence is difficult in existing systems.



Disadvantages of Existing System

- ❖ High processing time for claim verification.
- ❖ Increased chances of human error and missed fraud cases.
- ❖ Difficult to identify edited or manipulated images.
- ❖ Requires large manpower for claim inspection.
- ❖ Limited scalability for handling many claims.

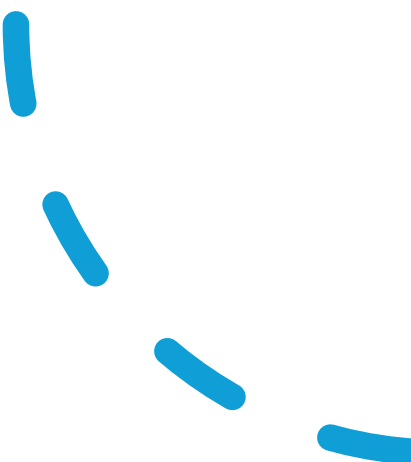
Proposed System

- The proposed system uses deep learning and image verification techniques for automatic fraud detection.
- YOLOv8 is used for object detection and identifying damaged vehicle parts.
- CNN models are used for image classification and tampering detection.
- Metadata analysis is performed to verify image authenticity.
- A fraud detection engine combines multiple indicators such as object mismatch, anomaly detection, and damage inconsistency.
- A Flask-based web application provides real-time claim analysis



Advantages of Proposed System



- ❖ Faster claim verification process.
 - ❖ Higher fraud detection accuracy.
 - ❖ Reduced manual effort and inspection cost.
 - ❖ Can detect fake, tampered, or inconsistent claim images.
 - ❖ Scalable and suitable for large insurance datasets.
 - ❖ Improves reliability and efficiency of insurance claim processing.
- 

5. Requirements

Software Requirements

- ❖ Operating System: Windows 7 Ultimate
- ❖ Coding Language: Python
- ❖ Front-end: Python(Tkinter)

Hardware Requirements

- ❖ Processor: Pentium IV
- ❖ RAM: 4GB (minimum)
- ❖ Hard disk: 20GB
- ❖ Keyboard: Standard Windows Keyboard
- ❖ Mouse: Two or Three Button Mouse
- ❖ Monitor: SVGA

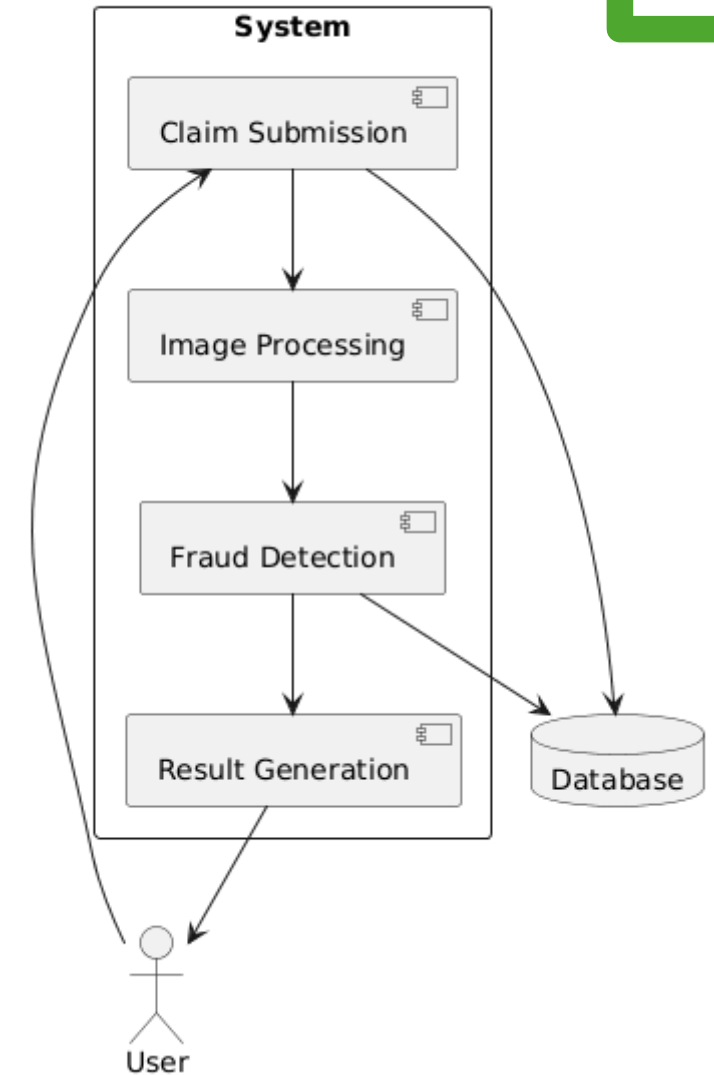
6. System Design and Methodology

Techniques/Algorithms Used:

- ❖ YOLOv8
- ❖ Convolutional Neural Network (CNN)
- ❖ OpenCV
- ❖ Anomaly Detection
- ❖ Metadata Analysis
- ❖ Flask

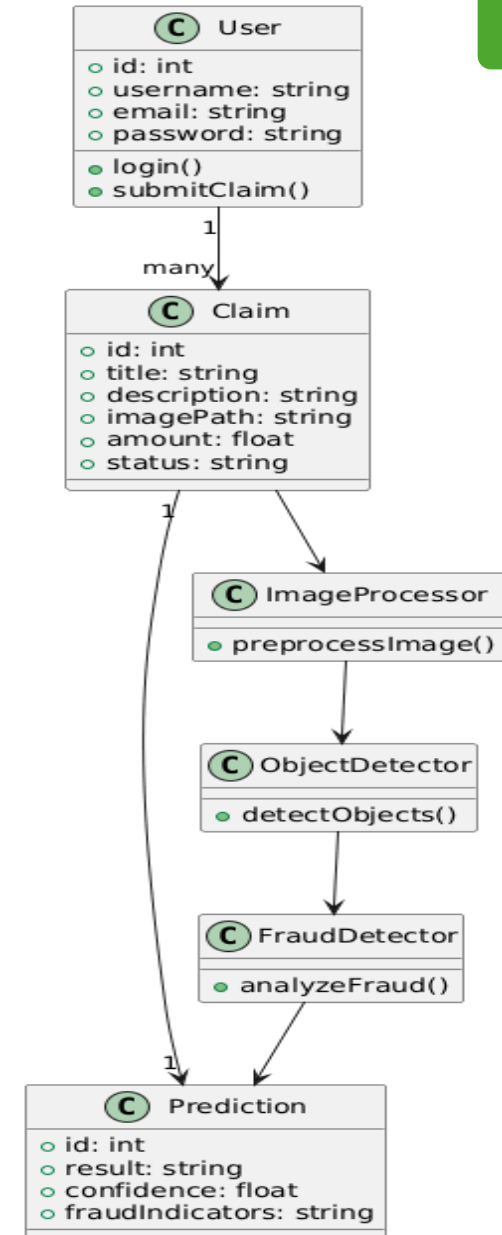
Data Flow Diagram

Data Flow Diagram (Level 1)

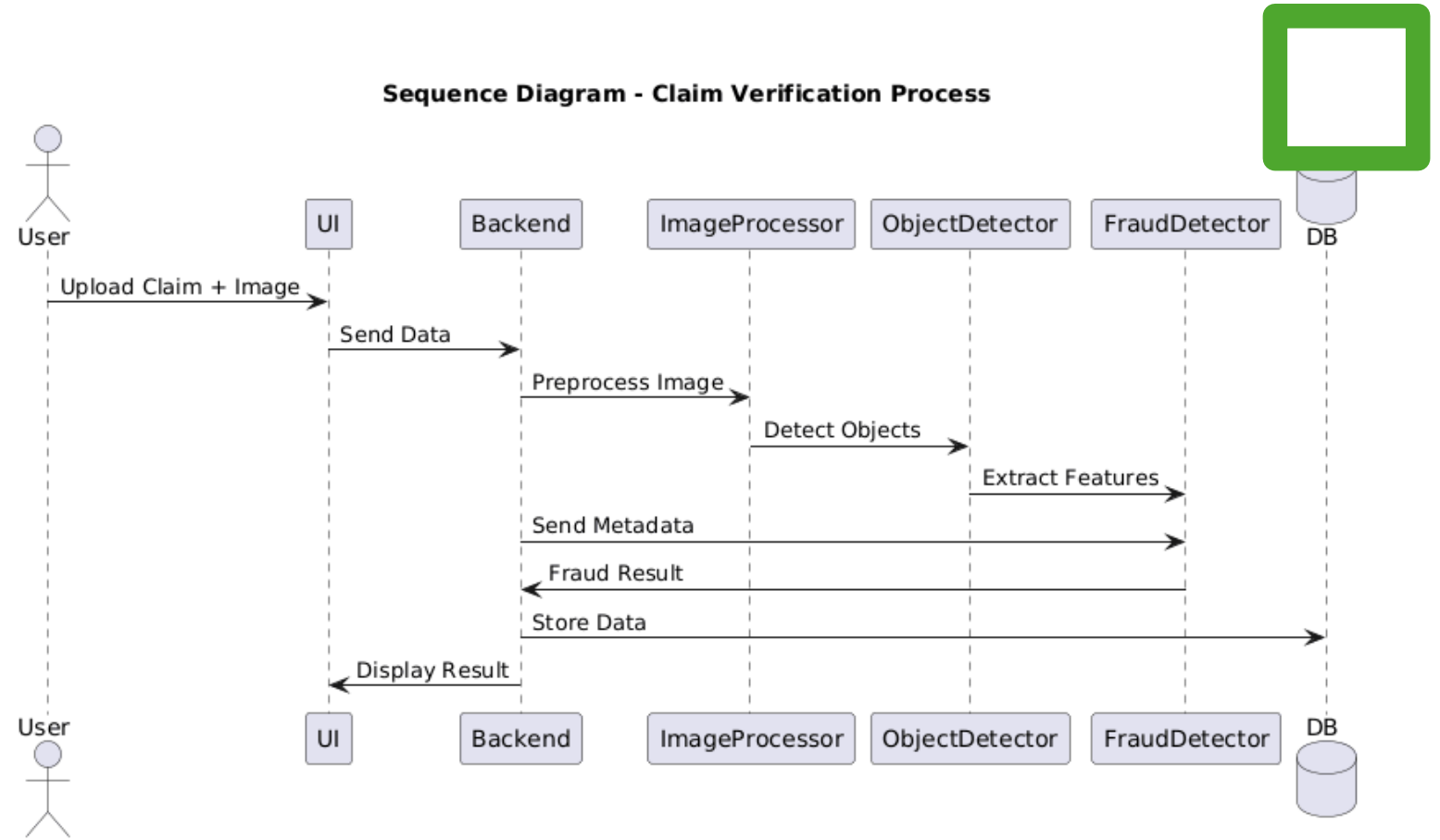


Class Diagram

Class Diagram - Insurance Claim Verification System

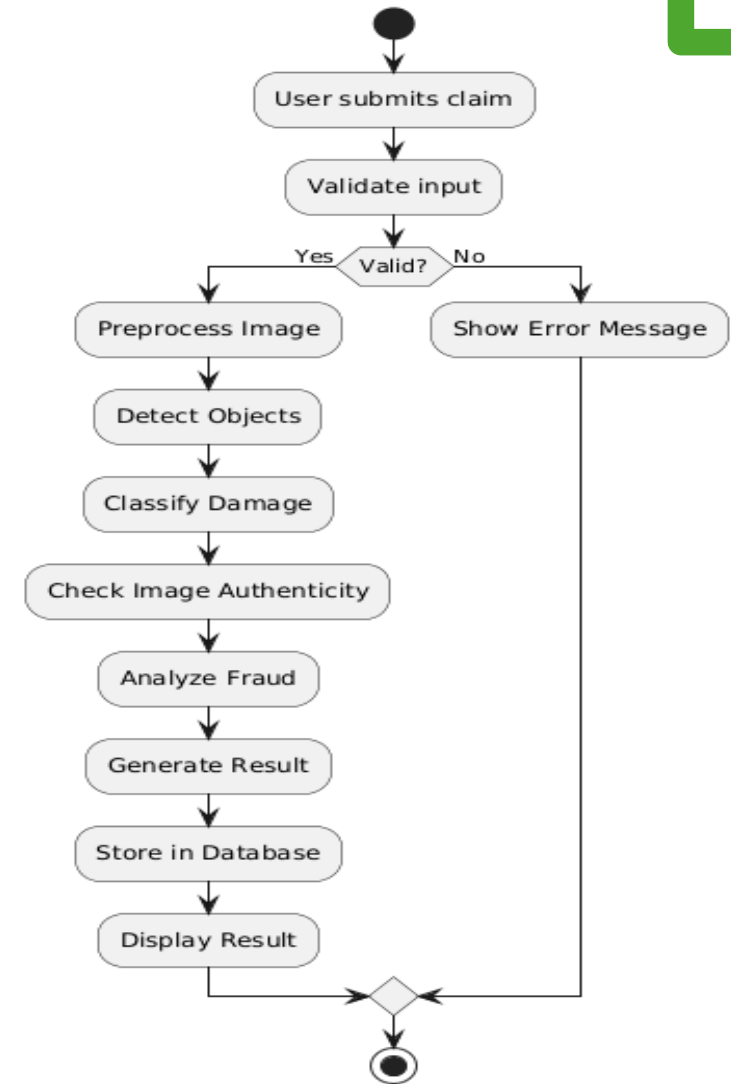


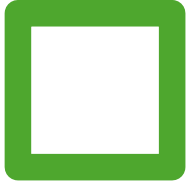
Sequence Diagram



Activity Diagram

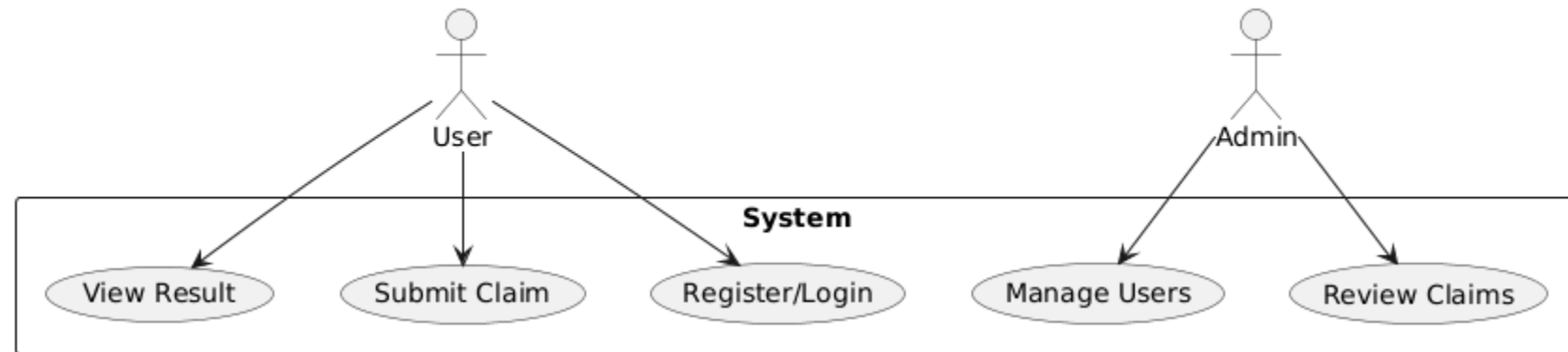
Activity Diagram - Claim Processing Workflow



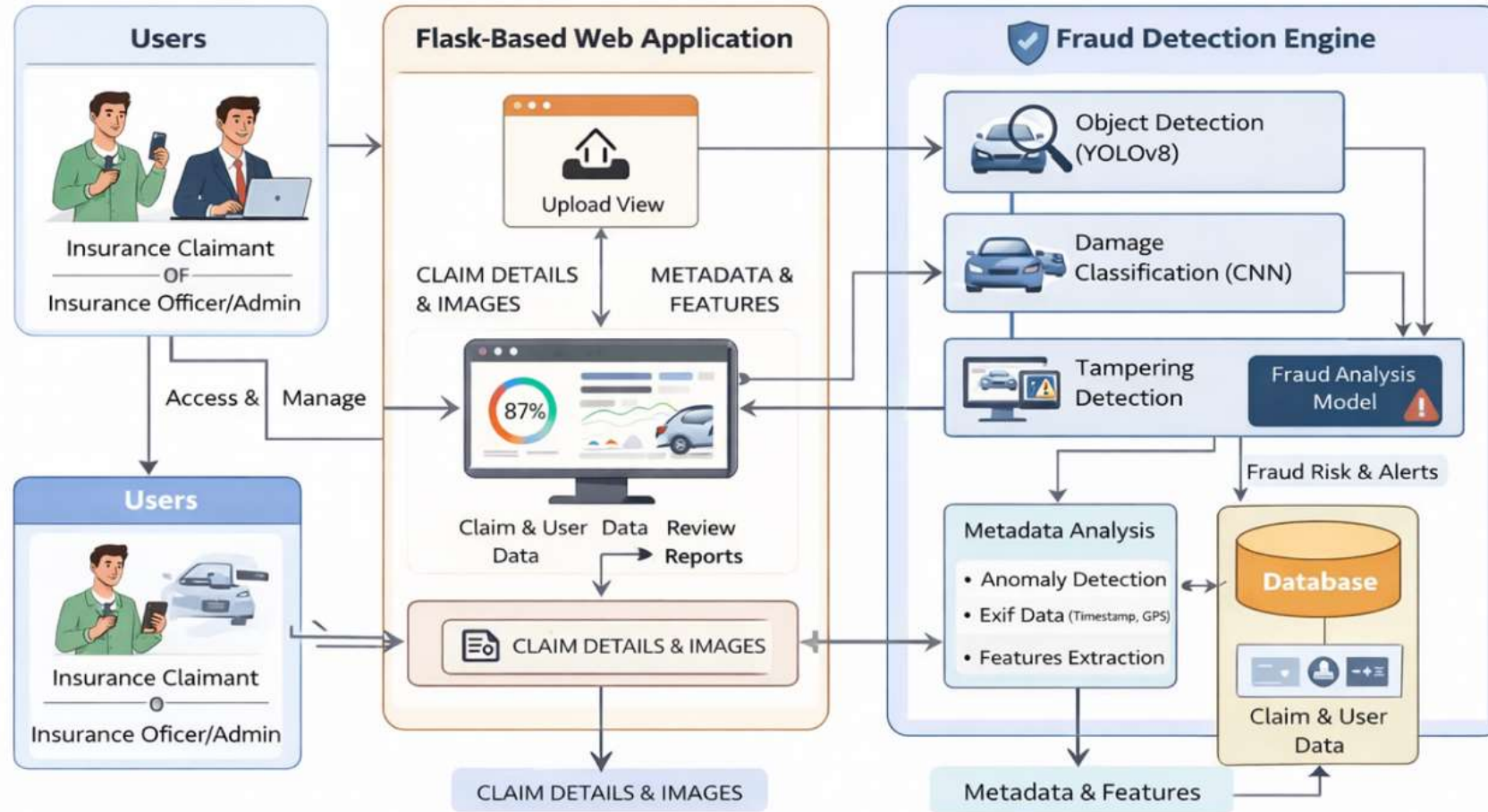


Use Case Diagram

Use Case Diagram - Insurance System



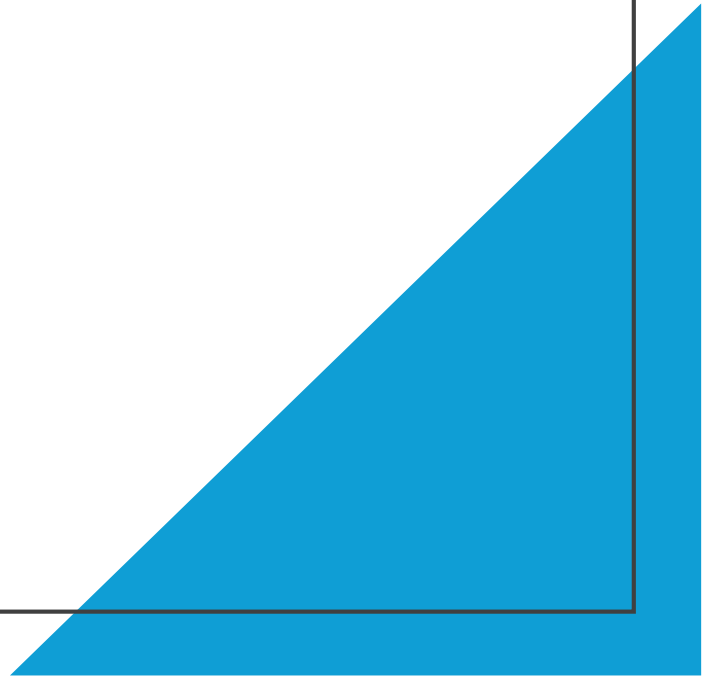
System Architecture



SYSTEM TESTING

- ❖ Unit Testing: Each module such as image upload, preprocessing, object detection, image classification, and metadata extraction is tested individually.
- ❖ Integration Testing: Ensures all modules work together correctly without errors.
- ❖ Functional Testing: Verifies that the system performs all required functions like uploading images, detecting damage, and classifying fraud.
- ❖ Performance Testing: Checks the response time and efficiency of the system when handling multiple claim images.
- ❖ Accuracy Testing: Measures how accurately the system identifies genuine and fraudulent claims.
- ❖ Usability Testing: Ensures the web application is simple, user-friendly, and easy to navigate.
- ❖ Security Testing: Verifies that user data and uploaded images are protected from unauthorized access.
- ❖ Validation Testing: Confirms that the predicted results match actual claim conditions and known fraud cases.

Output Screens





ShieldAI

[User Login](#)[Admin](#)[Get Started](#)

Powered by Hugging Face Transformers

Detect Insurance Fraud Instantly

Advanced AI vision analysis that identifies fraudulent claims with 94.7% accuracy. Upload. Analyze. Decide — in seconds.

[Start Free Analysis](#)[Admin Dashboard](#)



Welcome Back

Sign in to your ShieldAI account

✉ Email

you@example.com

🔒 Password

.....

🔑 Demo: john@example.com / password123

➔ Sign In

No account? [Sign Up](#) · [← Home](#)

Hello, john_doe! 🖐️

[+ Submit New Claim](#)

2 Total Claims

 0 Flagged
Fraud

2 Verified Real

 0 Pending

Recent Claims

Full History →

Side-swipe accident

📅 Mar 27, 2026 💰 \$ \$770.89

96.58% confidence

Hail storm damage

 john_doe
john@example.com



Admin Portal

Secure access to ShieldAI control center

Username

admin

Password

.....



Default: admin / admin123

→ Sign In as Admin

← Back to Home



ENG
IN



20:24
30-03-2026

3

Total Users

#	USER	EMAIL	CLAIMS	FRAUD DETECTED	JOINED	STATUS	ACTIONS
3	M mike_johnson	mike@example.com	4	2 fraud	Mar 30, 2026	Active	Delete
2	J jane_smith	jane@example.com	4	2 fraud	Mar 30, 2026	Active	Delete
1	J john_doe	john@example.com	2	Clean	Mar 30, 2026	Active	Delete

8. Conclusion

- ❖ The proposed system uses deep learning and computer vision to improve insurance claim verification.
- ❖ YOLOv8 and machine learning algorithms help detect fraud accurately and automatically.
- ❖ The system performs real-time image analysis and identifies inconsistencies in claim evidence.
- ❖ It combines image preprocessing, damage detection, authenticity verification, and fraud classification.
- ❖ The framework reduces manual effort, processing time, and operational costs.
- ❖ The modular design allows easy updates with new models and technologies.
- ❖ Overall, the system improves transparency, efficiency, reliability, and fairness in insurance claim processing.

9. Future Scope

- ❖ Future systems can use advanced deep learning models like Vision Transformers and multimodal architectures.
- ❖ Combining image analysis with text-based claim descriptions can improve fraud detection accuracy.
- ❖ Real-time video analysis can be added for live claim verification instead of only static images.
- ❖ Geolocation data and metadata analysis can help verify incident locations and image authenticity.
- ❖ Blockchain technology can provide secure and tamper-proof claim record storage.
- ❖ Reverse image search and external databases can help detect duplicate or reused claim images.
- ❖ Cloud-based deployment can improve scalability and accessibility for insurance companies.
- ❖ Continuous training with real-world datasets can further improve system accuracy and performance.